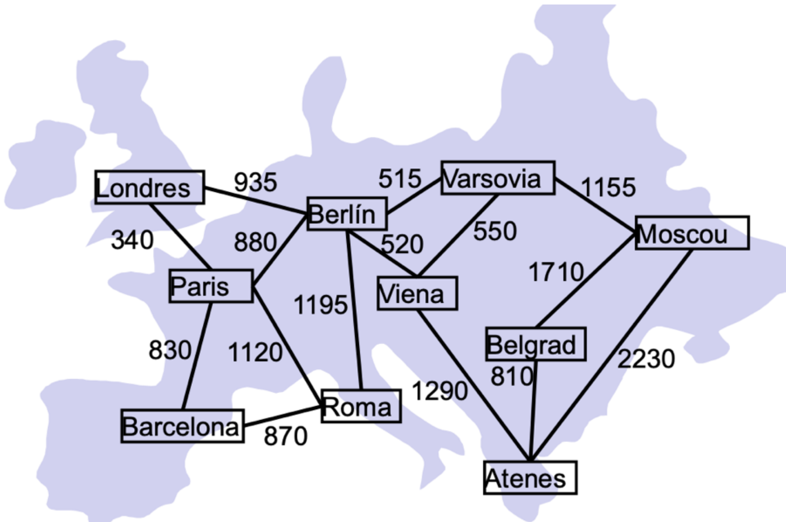
Piotr Bonar

Artificial Intelligence

Problems nr 4

Problem 1:

	<u>**Moscow**</u>
1-Barcelona	3015
2-Paris	2490
3-Londres	2505
4-Roma	2380
5-Berlin	1610
6-Viena	1670
7-Atenes	2230
8-Varsovia	1155
9-Belgrad	1710



Notation: [sum of g and h, [route], g, h]

Step 1.

[3015, [1], 0, 3015]

Step 2.

Extending: [3015, [1], 0, 3015]

[3250, [1,4], 870, 2380], [3320, [1,2], 830, 2490]

Step 3.

Extending: [3250, [1,4], 870, 2380]

[3320, [1,2], 830, 2490], [3675, [1,4,5], 2065, 1610], ~~[4480, [1,4,2], 1990, 2380],~~

~~[4755, [1,4,1], 1740, 3015]~~

Step 4.

Extending: [3320, [1,2], 830, 2490]

[3320, [1,2,5], 1710, 1610], ~~[3675, [1,4,5], 2065, 1610],~~ [3675, [1,2,3], 1170, 2505],

~~[4330, [1,2,4], 1950, 2380], [4675, [1,2,1], 1660, 3015]~~

Step 5.

Extending: [3320, [1,2,5], 1710, 1610]

[3380, [1,2,5,8], 2225, 1155], [3675, [1,2,3], 1170, 2505], [3900, [1,2,5,6], 2230, 1670],

~~[…, [1,2,5,4], …, …], […, [1,2,5,2], …, …], […, [1,2,5,3], …, …]~~

Step 6.

Extending: [3380, [1,2,5,8], 2225, 1155],

[3380, [1,2,5,8, Moscow], 3380, 0], [3675, [1,2,3], 1170, 2505], [3900, [1,2,5,6], 2230, 1670],

~~[…, [1,2,5,8,5], …, …], […, [1,2,5,8,6], …, …]~~

Problem 2:

Price per liter ***(P)*:**	0.75 euros / liter
Fuel usage per km ***(F)***:	12 liters / km
Minimum taxes per facility **(T)**:	6000 euros
Distance between A and B:	**D(A, B)**

$$Cost(A, B) = P * F * D(A, B) + T$$

$$h(n) = P * F * D(n, goal) + T$$

$$g(h) = \text{sum of all } Cost(A,B) \text{ from the route}$$

Problem 3:

b* from $h_1 \sim 1.5$ (best)
b* from $h_2 \sim 3$ (worst)
b* from $h_3 \sim 1.75$ (2^{nd})
b* from $h_4 \sim 2.5$ (3^{rd})

Problem 4:

b* from $h_5 \sim 2$ (worst)
b* from $h_6 \sim 1.75$ (3^{rd})
b* from $h_7 \sim 1.5$ (2^{nd})
b* from $h_8 \sim 1.25$ (best)

Problem 5:

Best to worst:
$h_8, h_1, h_7, h_6, h_3, h_5, h_4, h_2$